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Computational Finance Project
Mathematical Engineering - Quantitative Finance

Multi-Commodity Curves, PCA Calibration and Pricing

Group 3 - Project 2

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i) Data cleaning & statistical analysis

Dataset description and objective

We consider the dataset `power_DE_FR.csv`, containing daily time series of German and French electricity swap prices over approximately ten years. The contracts are labelled according to their delivery period:

- `Mc1`, `Mc2`, `Mc3`, ...: monthly delivery contracts, where `Mc1` delivers in the first delivery month after the observation date, `Mc2` in the following month, etc.
- `Qc1`, `Qc2`, ...: quarterly delivery contracts.
- `Yc1`, `Yc2`, ...: yearly delivery contracts.

Throughout the assignment we assume the valuation date is **5 November 2025**. The goal is to calibrate a market model for the joint German–French term structure and to price structured payoffs consistently with the modelling approach proposed in [1].

Handling missing values and contract selection

To deal with the fact that the raw dataset exhibits missing values (NaNs), we first applied the exclusions of `Yc3` **(both markets)** because it lies beyond the time horizon required for the subsequent analysis and is heavily affected by missing data, and `Mc4` **(both markets)** because its information can be incorporated/reconstructed through the quarterly constraint `Qc2` when building a coherent monthly curve. After these exclusions, remaining missing values are filled via **linear interpolation** along the time dimension for each contract series. This choice preserves continuity of the time series while avoiding model-implied assumptions prior to curve construction.

Statistical analysis

We computed the log-returns

$$r_t = \Delta \log F_t,$$

for each remaining contract series, and reported standard descriptive statistics: **sample mean, volatility (standard deviation), skewness and kurtosis**. Accordingly to the paper [1] we assessed stationarity of log-returns though the **ADF tests** and checked Gaussianity via **Jarque–Bera tests**. As expected (and anticipated in [1]), while the first test successfully allowed us to conclude that log-returns are actually stationary, the second one lead us to verify that log-returns' distribution is skewed, presents heavy tails and is peaked. In order to get a simple and stable calibration methodology we assume the hypothesis of normality since it is commonly accepted by practitioners. In order to mitigate the impact of extreme observations on the calibration and correlation analysis, a robust winsorization procedure was applied to log-returns. Following the approach adopted in [2], outliers were identified using an interquartile range (IQR) criterion: observations lying outside the interval

$$[Q_1 - k \cdot \mathrm{IQR},\ Q_3 + k \cdot \mathrm{IQR}],$$

with $k = 3$, were clipped to the corresponding boundary values. This procedure preserves the full time structure of the series, avoids discarding observations, and ensures numerical stability while retaining the relevant economic information contained in price spikes.

Table 1.1: Descriptive statistics of daily log-returns for power forward contracts

Contract	Mean	Std	Skew	Kurtosis
TRDEBMC1	3.683e-05	0.036701	0.0026837	5.2501
TRDEBMC2	0.00030899	0.032224	0.050345	5.1343
TRDEBMC3	0.00017696	0.028700	0.106690	5.0930
TRDEBQC1	0.00012483	0.029508	0.030380	5.1077
TRDEBQC2	0.00054507	0.025525	0.127120	5.3700
TRDEBQC3	0.00079144	0.022999	0.068244	5.5582
TRDEBQC4	0.00077987	0.021781	0.093665	5.4225
TRDEBYC1	0.00058180	0.022169	-0.0027955	5.2931
TRDEBYC2	0.00063079	0.017545	0.101430	5.5827
TRFRBMC1	-0.00079846	0.046190	0.097672	5.2359
TRFRBMC2	-0.00033519	0.037692	0.091636	5.2254
TRFRBMC3	7.2135e-05	0.026761	0.206790	5.0242
TRFRBQC1	-0.00041504	0.031719	0.108460	5.1520
TRFRBQC2	0.00027942	0.024577	-0.045108	4.9056
TRFRBQC3	0.00090565	0.020432	0.035893	4.9310
TRFRBQC4	0.00061562	0.015551	0.076026	4.6130
TRFRBYC1	0.00035883	0.020682	-0.0028083	5.1704
TRFRBYC2	0.00032405	0.014222	0.060815	4.9620

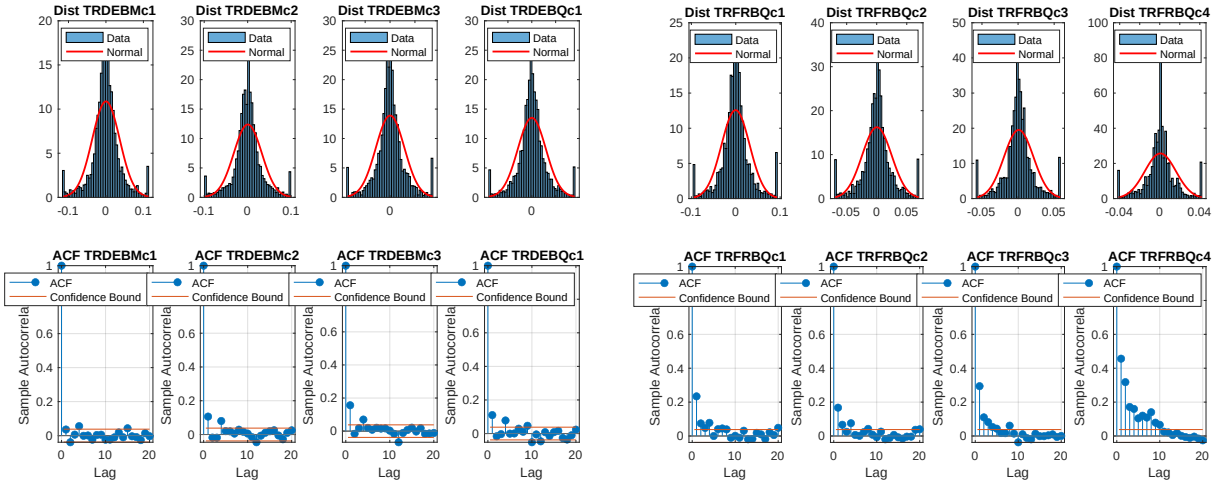


Figure 1.1: Distribution and autocorrelation (ACF) of daily log-returns for German and French power forward contracts. Results are reported for the first three monthly maturities and first quarterly one for the German market; while for the French market we report the quarterly contracts only. In general, a similar empirical behaviour is observed across all maturities.

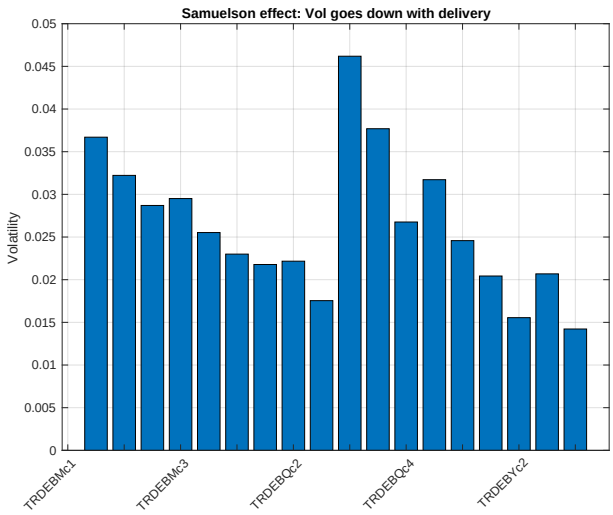


Figure 1.2: Samuelson effect for power forward contracts: volatility of daily log-returns as a function of time to maturity.

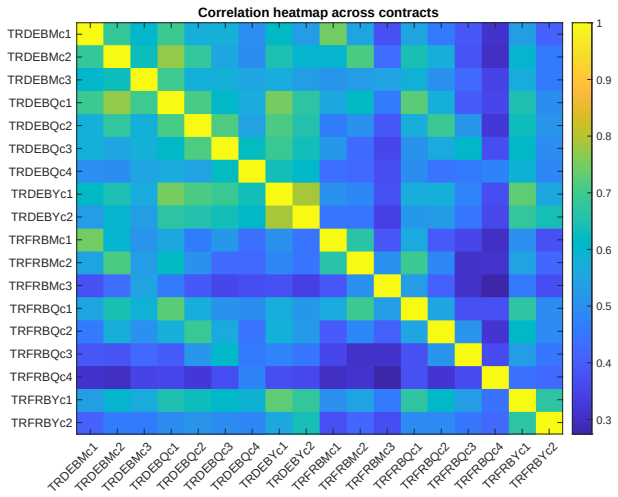


Figure 1.3: Correlation structure of daily log-returns for power forward contracts. Lighter cells indicate stronger positive correlation, highlighting common market drivers across maturities and regions.

ii) Monthly fwd curve construction

A flat monthly curve consistent with traded products

The dataset contains a mix of monthly, quarterly and yearly delivery swaps. Following [1], we construct a **monthly forward curve** (for each market) on a fixed horizon of **24 months**, represented as a step-wise (piecewise constant) structure across monthly buckets. The aim is to obtain synthetic monthly forwards:

$$\{F(t; M01), F(t; M02), \ldots, F(t; M24)\},$$

such that every observed market quote (monthly/quarterly/yearly) is matched by the appropriate average of the underlying monthly buckets (no-arbitrage consistency).

No-arbitrage linear constraints

For each observation date and for each market separately, we define a vector of unknown monthly forwards

$$x = (M01, \ldots, M24)^\top.$$

We then impose linear constraints of the form

$$Ax \approx b,$$

where each row corresponds to a traded product:

- **Monthly contracts**: `Mc1-Mc3` are mapped directly to $M01, M02, M03$ as equality constraints (hard constraints).
- **Quarterly contracts**: each `Qc`q equals the average of the three monthly buckets spanning that quarter (weights 1/3, ACT360).
- **Yearly contracts**: each `Yc`y equals the average of the twelve monthly buckets spanning that calendar year (weights 1/12, ACT360).

Rolling delivery mapping

A key implementation detail is the **rolling convention** for continuation contracts:

- `Mc1` always refers to the first delivery month after the observation date.
- `Qc1` refers to the *first quarter not yet started* relative to the observation date (quarter buckets: Jan–Mar, Apr–Jun, Jul–Sep, Oct–Dec). Hence, if the observation date lies in Jan/Feb/Mar, then `Qc1` maps to Apr–Jun; if in Apr/May/Jun then `Qc1` maps to Jul–Sep, etc.
- `Yc1` refers to the *first calendar year not yet started*; for a date in year Y, `Yc1` maps to Jan–Dec of year $Y + 1$.

This logic was implemented by computing, for each date, the start index of the delivery window of Qc1 and Yc1, and then shifting subsequent quarters/years accordingly.

Numerical resolution and flat forward completion

For each observation date, the system of no–arbitrage constraints described in Section 4.1.1 of [1] leads to a linear system of the form

$$Ax = b,$$

where $x \in \mathbb{R}^H$ represents the monthly forward prices over a fixed horizon H, A collects the aggregation constraints coming from monthly, quarterly and yearly contracts, and b contains the corresponding observed market prices.

Due to the rolling structure of traded contracts, the matrix A is in general rank–deficient: only a subset of monthly forwards is directly constrained by market quotes at a given date. To address this issue, we proceed as follows.

First, we identify the subset of *active months*, namely those months that effectively appear in at least one constraint:

$$\mathcal{I} = \{m \in \{1, \dots, H\} : \exists\, k \text{ such that } A_{k,m} \neq 0\}.$$

The reduced system

$$A_{\mathcal{I}} x_{\mathcal{I}} = b$$

is then solved in the least–norm sense using a minimum–norm least–squares approach. This choice ensures numerical stability and avoids introducing spurious curvature in underdetermined directions.

Second, since the remaining months are not directly constrained by market data, we complete the curve by imposing a *flat forward extrapolation*. More precisely, once the first valid monthly forward price is identified, all preceding months are backward–filled with this value, while all subsequent unconstrained months are forward–filled using the last available value. This procedure guarantees a continuous and economically reasonable forward curve, consistent with the idea of a piecewise–constant monthly structure.

This construction yields a unique, well–defined monthly forward curve at each observation date, which is fully consistent with the aggregation constraints of the observed futures and suitable for subsequent statistical analysis and Monte Carlo simulation.

Statistical analysis of synthetic monthly forwards

After constructing the synthetic monthly curves (German and French), we repeat the same return-based statistical analysis on the synthetic monthly forward time series:

- descriptive statistics and tests on $r_t = \Delta \log M_t$,
- correlation structure across synthetic maturities,
- verification of Samuelson effect on the monthly grid.

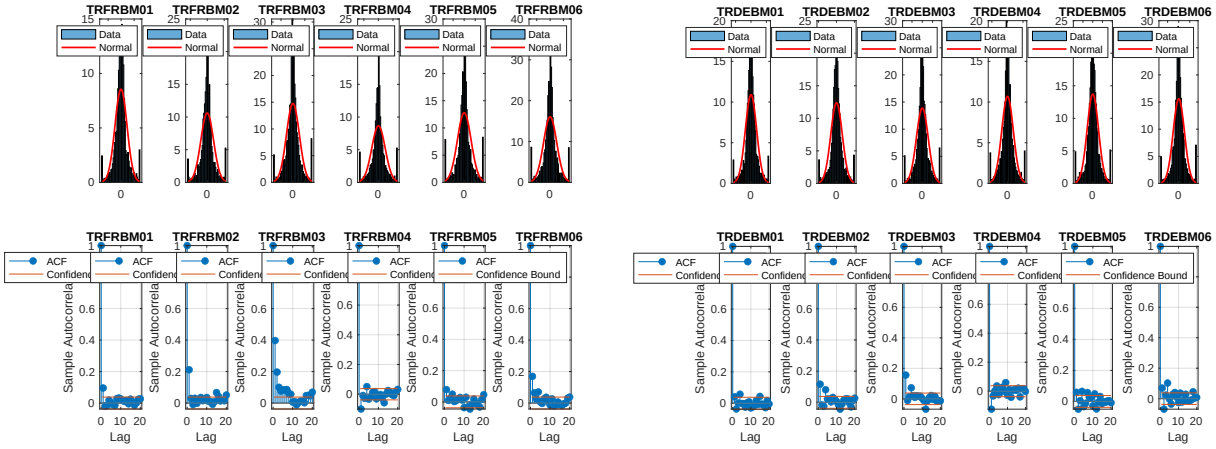


Figure 1.4: Distribution and autocorrelation (ACF) of daily log-returns for German and French power forward contracts. Results are reported for the first 6 monthly maturities for both German and French market. In general, a similar empirical behaviour is observed across all maturities.

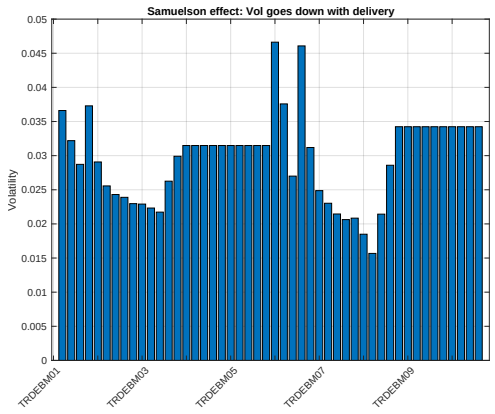


Figure 1.5: Samuelson effect for power forward contracts: volatility of daily log-returns as a function of time to maturity.

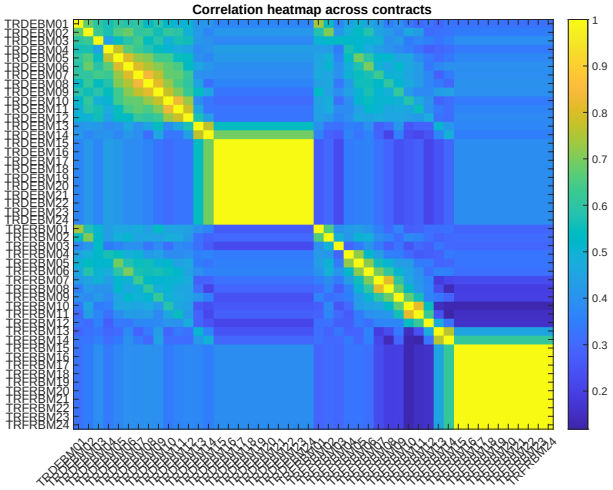


Figure 1.6: Correlation structure of daily log-returns for power forward contracts. Lighter cells indicate stronger positive correlation, highlighting common market drivers across maturities and regions.

iii) PCA factor reduction on forward returns

Motivation

The model in [1] relies on representing the joint dynamics of a large set of forward returns through a smaller number of common risk drivers. Empirically, electricity curves exhibit strong co-movement across maturities (and across related markets such as DE and FR), suggesting that a low-dimensional factor model can capture most of the variance.

PCA setup

We apply PCA to the matrix of synthetic forward log-returns:

$$R \in \mathbb{R}^{T \times d},$$

where d is the number of synthetic monthly forwards across both markets (German 24 + French 24 = 48) and T is the number of time observations. PCA returns:

- `coeff`: eigenvectors (loadings) defining principal directions in the space of contract returns,
- `score`: projected time series of the factors (principal component realizations),
- `latent`: eigenvalues, i.e. variance explained by each component,
- `explained`: percentage of total variance explained by each component.

Choice of retained variance and number of factors

We select the number of factors m such that the cumulative explained variance exceeds a chosen threshold:

$$\sum_{i=1}^{m} \text{explained}_i \geq \gamma,$$

with $\gamma = \mathbf{95\%}$ (chosen as a balance between dimensionality reduction and accuracy). By doing so we retains the vast majority of the empirical covariance structure used for Monte Carlo pricing, we keep the factor dimension small enough to be computationally efficient, we are consistent with the strong co-movement typically observed in power forward curves.

Table 1.2: Number of principal components required to explain a given fraction of total variance.

Explained variance threshold	**Number of factors**
95%	14

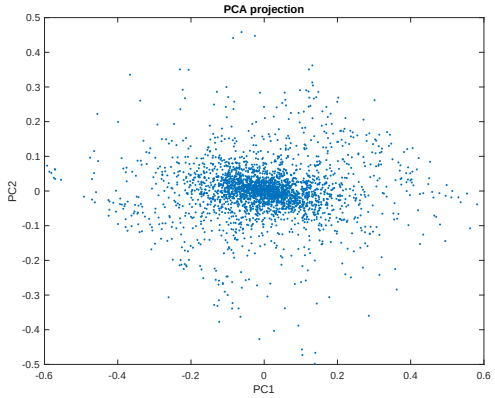


Figure 1.7: Projection of daily log-returns onto the first two principal components.

The cloud of observations in Figure 1.7 is concentrated around the origin, indicating that the dominant sources of variability are captured by a small number of common factors. The absence of pronounced clustering or directional patterns suggests a relatively stable correlation structure over time, with the first two components mainly reflecting broad market-wide movements rather than contract-specific idiosyncratic behaviour.

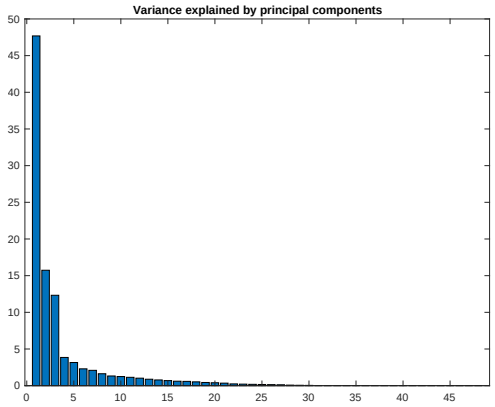


Figure 1.8: Explained variance ratio of principal components.

The first few components account for a substantial share of the overall variability, while the contribution of higher-order components rapidly decreases. This evidence confirms the presence of strong common drivers across power forward contracts and supports the use of low-dimensional factor representations for describing the joint dynamics of returns.

iv) Pricing of the Chooser Option

Setup and payoff

In this section we price a chooser option written on two monthly power forwards, one referring to the German market and one to the French market, both delivering in April 2027. The option expires on February 5, 2027 and pays at expiry the maximum between the two forward prices. Let F_t^{DE} and F_t^{FR} denote the German and French forward prices observed at time t, referring to the same delivery period. The payoff of the chooser option at maturity T is given by

$$\Pi_T = \max\left(F_T^{DE},\, F_T^{FR}\right).$$

The price of the option at the valuation date t_0 is therefore

$$V_0 = DF(t_0, T)\,\mathbb{E}\big[\max\left(F_T^{DE},\, F_T^{FR}\right)\big],$$

where $DF(t_0, T)$ is the discount factor between t_0 and T, taken from market data.

Stochastic modeling of forward prices

Following the framework of the paper, forward prices are modeled under the risk–neutral measure as lognormal processes driven by a finite number of Brownian factors. In differential form,

$$\frac{dF_t^i}{F_t^i} = \boldsymbol{\sigma}_i(t) \cdot d\mathbf{W}_t, \qquad i \in \{\mathrm{DE}, \mathrm{FR}\},$$

where $\boldsymbol{\sigma}_i(t)$ is a deterministic volatility vector.

Monte Carlo pricing via PCA

The volatility structure is calibrated using a Principal Component Analysis (PCA) on the historical daily returns of the synthetic forward curves. Retaining the minimum number of factors explaining more than 95% of the total variance, we obtain a low-dimensional representation of the covariance structure of forward returns. This allows us to reconstruct the joint distribution of the German and French forwards at the option maturity.

As a first approach, the chooser option is priced via Monte Carlo simulation. Let F_0^{DE} and F_0^{FR} denote the forward prices observed at the valuation date t_0. For each simulation, the terminal log-returns are generated as

$$R_T = Z\,\sqrt{\Lambda\, T}\,B^{\top},$$

where Z is a matrix of independent standard normal variables, B contains the PCA loadings, Λ is the diagonal matrix of retained eigenvalues, and T is the time to maturity expressed in business days.

Under the lognormal assumption, the terminal forward prices are obtained as

$$F_T^{DE} = F_0^{DE} \exp\!\left(-\tfrac{1}{2}\sigma_{DE}^2 T + R_T^{DE}\right),$$

$$F_T^{FR} = F_0^{FR} \exp\left(-\tfrac{1}{2}\sigma_{FR}^2 T + R_T^{FR}\right),$$

where the drift terms ensure the martingale property of the forwards.

The option price is then estimated as

$$V_0^{MC} = DF(t_0, T)\,\frac{1}{N}\sum_{k=1}^{N} \max\left(F_T^{DE,(k)},\, F_T^{FR,(k)}\right).$$

Lower and upper bounds are also computed:

$$V_0^{LB} = DF(t_0, T)\,\max(F_0^{DE}, F_0^{FR})$$

$$V_0^{UB} = DF(t_0, T)\,\left(F_0^{DE} + F_0^{FR}\right),$$

providing a consistency check for the Monte Carlo estimate.

Pricing method	**Option value**
Monte Carlo Estimate	105.1352
Lower bound	94.4766
Upper bound	163.5600

Table 1.3: Chooser option pricing: Monte Carlo estimate and theoretical bounds

Analytical decomposition of the payoff

Although the chooser option is priced numerically via Monte Carlo simulation, in this specific setting an explicit valuation formula can be derived under the lognormal assumption on forward prices. This provides both analytical insight and a useful benchmark for the numerical results.

The payoff of the chooser option can be rewritten using the elementary identity

$$\max(x, y) = x + \max(y - x, 0),$$

which holds for any real numbers x and y. Applying this identity to the terminal forward prices yields

$$\Pi_T = \max\left(F_T^{DE},\, F_T^{FR}\right) = F_T^{DE} + \max\left(F_T^{FR} - F_T^{DE},\, 0\right).$$

This representation shows that the chooser option can be interpreted as the sum of a long position in the German forward and an option written on the spread between the French and German forwards. Since forward prices are martingales under the risk–neutral measure, the discounted expectation of the first term is simply equal to its current value F_0^{DE}. The pricing problem therefore reduces to the evaluation of the option component

$$\mathbb{E}\left[\max\left(F_T^{FR} - F_T^{DE},\, 0\right)\right].$$

The spread payoff can be further factorized by extracting one of the forward prices:

$$\max\left(F_T^{FR} - F_T^{DE}, 0\right) = F_T^{DE}\left(\frac{F_T^{FR}}{F_T^{DE}} - 1\right)^+.$$

This factorization is crucial, as it isolates a strictly positive tradable asset F_T^{DE} from a payoff depending only on the ratio of the two forward prices.

Change of numeraire and closed-form valuation

Consider now the change of numeraire associated with the German forward price F_t^{DE}. By the change–of–numeraire theorem, for any integrable payoff X_T one has

$$\mathbb{E}^{\mathbb{Q}}\left[F_T^{DE} X_T\right] = F_0^{DE}\,\mathbb{E}^{\mathbb{Q}^{DE}}[X_T],$$

where $\mathbb{Q}^{DE}$ denotes the probability measure associated with the numeraire F_t^{DE}.

Applying this result with

$$X_T = \left(\frac{F_T^{FR}}{F_T^{DE}} - 1\right)^+,$$

the expectation of the spread payoff becomes

$$\mathbb{E}\left[\max\left(F_T^{FR} - F_T^{DE}, 0\right)\right] = F_0^{DE}\,\mathbb{E}^{\mathbb{Q}^{DE}}\left[\left(\frac{F_T^{FR}}{F_T^{DE}} - 1\right)^+\right].$$

Under the lognormal dynamics assumed for the forward prices, the ratio

$$X_t = \frac{F_t^{FR}}{F_t^{DE}}$$

follows a lognormal process under the measure $\mathbb{Q}^{DE}$. As a consequence, the remaining expectation corresponds to the payoff of a call option with unit strike written on a lognormal underlying, which admits a closed-form valuation.

Denoting by $N(\cdot)$ the standard normal cumulative distribution function and defining

$$d_1 = \frac{\log\left(F_0^{FR}/F_0^{DE}\right) + \frac{1}{2}\sigma_X^2 T}{\sigma_X\sqrt{T}},$$

$$d_2 = d_1 - \sigma_X\sqrt{T},$$

where

$$\sigma_X^2 = \sigma_{FR}^2 + \sigma_{DE}^2 - 2\rho\,\sigma_{FR}\sigma_{DE},$$

the closed-form value of the chooser option is finally given by

$$V_0^{CF} = DF(t_0, T)\left[F_0^{DE} + F_0^{FR} N(d_1) - F_0^{DE} N(d_2)\right].$$

This expression coincides with the classical closed-form valuation formula for exchange options, commonly referred to as the Margrabe formula.

Pricing method	**Option value**
Closed-form (Margrabe)	105.1299

Table 1.4: Chooser option pricing: closed-form valuation via Margrabe formula

The closed-form price is found to be in excellent agreement with the Monte Carlo estimate. While Monte Carlo simulation is not strictly necessary for pricing the chooser option, it provides a flexible framework that can be readily extended to more complex payoffs for which no explicit solution is available.

v) Pricing of the Daily Swing Option

Setup and Option Characteristics

In this final section, we price a **Daily Swing Option** on the German electricity spot market for the delivery period of November 2027 ($T = 30$ days). Consistent with the project requirements, we consider only **upswings** (rights to buy) with the following parameters:

- **Delivery Period**: 1 Nov 2027 to 30 Nov 2027.
- **Strike Price (K)**: 40 /MWh.
- **Total Exercise Rights (N)**: Base case $N = 15$.
- **Valuation Date**: 5 November 2025.

Spot Price Dynamics and Path Simulation

Following Section 3.4 of [1], the spot price S_t is simulated using the PCA-calibrated HJM framework. For each Monte Carlo path j, the spot price for a delivery day d is:

$$S_d^{(j)} = F(0, T_d) \exp\left(\sum_{i=1}^{m} \text{shock}_{i,d}^{(j)} - \frac{1}{2}\text{Var}_{total,d}\right)$$

where the total variance and shocks are derived from the 24-month synthetic forward curve and the identified risk factors.

Technical focus: the Longstaff-Schwartz algorithm for Swing Options

The valuation of a Daily Swing option is an optimal stochastic control problem. Unlike a standard American option, where the exercise ends the contract's life, here the holder possesses N rights that can be exercised over T days. This creates a multi-layered American-style problem.

At each daily step $t \in \{1, \ldots, 30\}$, and for each remaining number of rights $n \in \{1, \ldots, N\}$, the holder must decide whether to exercise a right based on the Bellman equation:

$$V(t, n) = \max\left(\pi_t + DF \cdot \mathbb{E}_t[V(t+1, n-1)],\ DF \cdot \mathbb{E}_t[V(t+1, n)]\right)$$

where $\pi_t = \max(S_t - K, 0)$ is the immediate payoff. The core difficulty lies in estimating the conditional expectation $\mathbb{E}_t[\cdot]$, which represents the **continuation value**.

We implement the **Longstaff-Schwartz (2001)** approach, which approximates this expectation using a least-squares regression on the simulated spot paths. Specifically:

1. **Basis Functions**: For each time t, we regress the realized future payoffs onto a set of basis functions of the current state S_t. In our code, we use a polynomial basis: $C(S_t) \approx \sum_{k=0}^{2} a_k (S_t)^k$.
2. **Regression**: The regression is performed only on "in-the-money" paths to increase

numerical stability, as these are the only paths where the exercise decision is non-trivial.

3. **Backward Induction**: We move backwards from T to t_0. For each path, if the immediate payoff π_t is greater than the estimated continuation value $\hat{C}(S_t)$, we record an exercise for that specific path and decrement the available rights n.

Convergence and Numerical Consistency

The algorithm was executed with 10^5 Monte Carlo paths. As shown in Figure 1.9, the option price follows a remarkably linear progression between the Lower Bound and the Upper Bound. Specifically, our estimate for $N = 15$ (**879.44€**) sits at 50.5% of the total potential value ($N = 30$), indicating that the marginal value of each additional "swing" remains almost constant in this specific calibration.

This quasi-linear behavior is characteristic of markets with high spot volatility and frequent price spikes: the "scarcity" of exercise rights is mitigated by the abundance of attractive exercise opportunities. From a numerical standpoint, this stability confirms that the Longstaff-Schwartz regressor, utilizing a polynomial basis, effectively captures the exercise boundary without premature saturation. It is worth noting that in such high-volatility regimes, the choice of the regression's basis functions becomes critical: a low-order polynomial is sufficient here to distinguish the "spike" states from the mean-reverting levels, ensuring a robust estimation of the continuation value across all N levels.

Numerical Results and Consistency Checks

The results for the base case and the theoretical bounds are reported in Table 1.5.

Metric	**Value ()**
Swing Price ($N = 15$)	879.442 906
Lower Bound ($N = 1$)	62.852 176
Upper Bound ($N = 30$)	1677.633 397

Table 1.5: Swing option pricing results (Only Upswings) via Longstaff-Schwartz.

The evolution of the Swing option price as a function of the exercise rights N is illustrated in Figure 1.9.

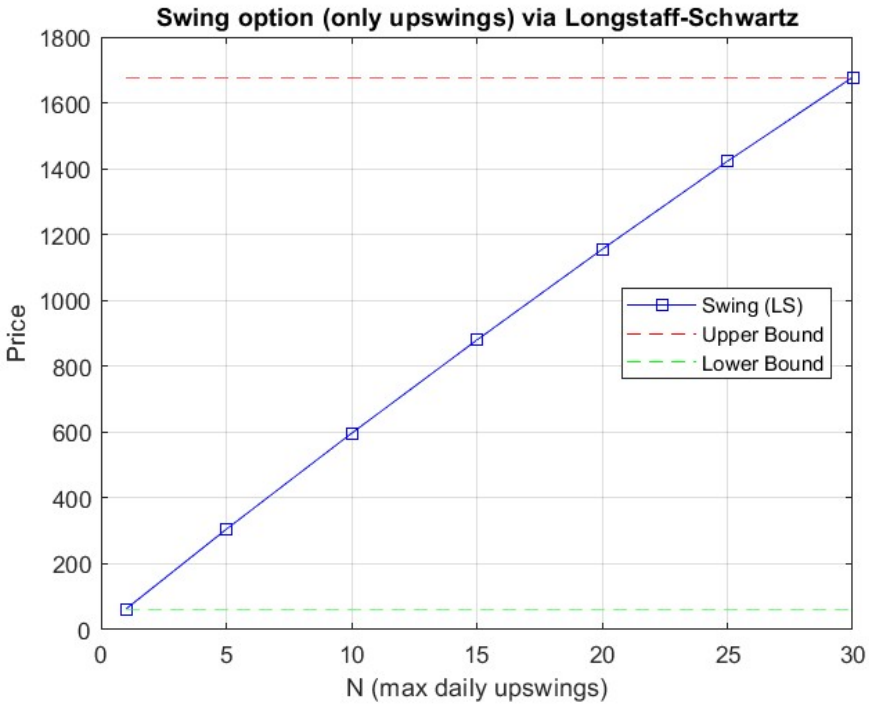


Figure 1.9: Swing option price as a function of the number of exercise rights N.

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[1] Gardini, A. and Santilli, M. (2023). *A Heath-Jarrow-Morton framework for energy markets: a pragmatic approach.*

[2] Baviera, R. and Santagostino, T. (2019). *Stop-loss and leverage in optimal statistical arbitrage with an application to energy market.*